FACTORY ACCEPTANCE TEST REPORT

Robotic Drilling and Fastening Cells 1 to 8 - Varnholt Aerostructures Limited, Sinfin Lane Works, Derby

Document no.: QRL-VAR-FAT-001 | Revision: C | Date: 02/08/2024
Supplier: Quellcrest Robotics Ltd, Unit 7, Tolland Park, Brackenhurst Road, Coventry CV3 4RZ
Customer PO: VAR-PO-23-0412 and VAR-PO-23-0412/A1

Revision history

Rev	Date	Description	Prepared	Approved
A	08/12/2023	FAT of Cells 1, 2 and 3	P. Sandhu	G. Whitlow
B	21/06/2024	Adds FAT of Cells 4, 5 and 6	P. Sandhu	G. Whitlow
C	02/08/2024	Adds FAT and re-test of Cells 7 and 8; NCR-031 and NCR-033 closed	P. Sandhu	G. Whitlow / B. Kowalczyk

1. Purpose and scope

1.1 This report records the factory acceptance tests ("FAT") of the eight robotic drilling and fastening cells supplied to Varnholt Aerostructures Limited ("Varnholt" or "the Buyer") under the Supply and Installation Agreement dated 14 March 2023, as varied by Variation Agreement No. 1 dated 9 January 2024 (Cells 7 and 8), clause 8 and Schedule 1 Part C.

1.2 All FATs were carried out in Test Bay T2 at the Supplier's Coventry facility. Each FAT was witnessed by the Buyer's Quality Lead, Mr T. Wierzbicki. The Buyer's Contract Manager, Ms R. Thorne, also attended the FATs of Cells 1, 4, 7 (first test and re-test) and 8 (re-test).

1.3 This report does not cover site acceptance tests (SAT), which are recorded separately on form QRL-F-SAT-02.

2. References

2.1 Supply and Installation Agreement, clause 8 (Testing) and Schedule 1 Part B (performance requirements) and Part C (Test Specification).

2.2 Variation Agreement No. 1, Annex (Cells 7 and 8: B-series fixtures, 14 m track, three fastener diameters).

2.3 Test procedure QRL-TP-117 rev 3; laser tracker calibration certificate CAL-LT-2023-0912 (valid to 30/09/2024).

2.4 Non-conformance reports NCR-031 (Cell 7) and NCR-033 (Cell 8).

3. Test set-up

3.1 Test coupon: 2024-T3 aluminium alloy, 6 mm + 4 mm stack, 50 holes per coupon (4.8 mm and 6.35 mm; for Cells 7 and 8 also 4.0 mm), drilled, countersunk and fastened by the Cell under test.

3.2 Positional accuracy measured as the true position of each hole against nominal, by laser tracker; the result reported is the worst hole on the coupon. Repeatability measured over 30 repeat cycles to a reference hole. Cycle time is the mean over 120 consecutive fasteners (drill, countersink, insert, swage).

3.3 Ambient temperature in T2 was held at 20°C ± 2°C during all tests.

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4. Acceptance criteria (Schedule 1 Part B)

Ref	Parameter	Requirement
B1	Positional accuracy	≤ 0.15 mm
B2	Repeatability	≤ ±0.05 mm
B3	Hole diameter	H8
B4	Countersink depth	±0.05 mm
B5	Cycle time per fastener	≤ 22.0 s
B6	Swage collar quality	zero defects
B8	Noise at 1 m	≤ 80 dB(A)

B7 (availability over an 8-hour run) is a SAT-only requirement.

5. Summary of results

Cell	FAT date	Attempt	Positional accuracy (mm)	Repeatability (mm)	Cycle time (s)	Result	Re-test	FAT Certificate
1	06/11/2023	1	0.11	±0.03	19.4	PASS	-	FATC-01
2	20/11/2023	1	0.12	±0.03	19.7	PASS	-	FATC-02
3	04/12/2023	1	0.10	±0.02	19.5	PASS	-	FATC-03
4	11/03/2024	1	0.12	±0.04	19.9	PASS	-	FATC-04
5	03/06/2024	1	0.13	±0.03	20.1	PASS	-	FATC-05
6	17/06/2024	1	0.11	±0.03	19.8	PASS	-	FATC-06
7	01/07/2024	1	0.21	±0.04	20.6	FAIL	22/07/2024	-
7	22/07/2024	2	0.12	±0.03	20.2	PASS	-	FATC-07
8	08/07/2024	1	0.19	±0.04	20.4	FAIL	29/07/2024	-
8	29/07/2024	2	0.13	±0.03	20.0	PASS	-	FATC-08

Cells 1 to 6 passed FAT at the first attempt. Cells 7 and 8 failed their first FAT on positional accuracy only (0.21 mm and 0.19 mm respectively, against the requirement of ≤ 0.15 mm) and passed on re-test. All other parameters were within limits at every attempt.

6. Results by Cell

Cell 1 - Bay 2

FAT date	Positional accuracy (mm)	Cycle time (s)	Result	Re-test
06/11/2023	0.11	19.4	PASS	None required

Hole diameter, countersink depth and swage collars: all conforming (0 defects / 120). Noise 76 dB(A).

Cell 2 - Bay 2

FAT date	Positional accuracy (mm)	Cycle time (s)	Result	Re-test
20/11/2023	0.12	19.7	PASS	None required

All other parameters conforming. Noise 77 dB(A).

Cell 3 - Bay 2

FAT date	Positional accuracy (mm)	Cycle time (s)	Result	Re-test
04/12/2023	0.10	19.5	PASS	None required

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All other parameters conforming. Noise 76 dB(A).

Cell 4 - Bay 3

FAT date	Positional accuracy (mm)	Cycle time (s)	Result	Re-test
11/03/2024	0.12	19.9	PASS	None required

All other parameters conforming. Noise 78 dB(A). Cell 4 was held in Coventry after FAT pending Delivery to Bay 3 (delivered 30/04/2024).

Cell 5 - Bay 3

FAT date	Positional accuracy (mm)	Cycle time (s)	Result	Re-test
03/06/2024	0.13	20.1	PASS	None required

All other parameters conforming. Noise 77 dB(A). FAT re-planned from April 2024 following late delivery of the HSK-63 spindle drive units by the Supplier's sub-supplier (units received 20/05/2024).

Cell 6 - Bay 3

FAT date	Positional accuracy (mm)	Cycle time (s)	Result	Re-test
17/06/2024	0.11	19.8	PASS	None required

All other parameters conforming. Noise 77 dB(A). Spindle drive units as Cell 5.

Cell 7 - Bay 2 (south)

FAT date	Positional accuracy (mm)	Cycle time (s)	Result	Re-test
01/07/2024	0.21	20.6	FAIL (B1)	22/07/2024
22/07/2024	0.12	20.2	PASS	-

First test: 9 of 50 holes outside 0.15 mm true position, worst 0.21 mm, all in the last 4 m of track travel. NCR-031 raised. Re-test within 15 Business Days per clause 8.4.

Cell 8 - Bay 2 (south)

FAT date	Positional accuracy (mm)	Cycle time (s)	Result	Re-test
08/07/2024	0.19	20.4	FAIL (B1)	29/07/2024
29/07/2024	0.13	20.0	PASS	-

First test: 6 of 50 holes outside 0.15 mm, worst 0.19 mm, same zone as Cell 7. NCR-033 raised. Re-test within 15 Business Days per clause 8.4.

7. Non-conformances

7.1 NCR-031 (Cell 7) and NCR-033 (Cell 8). Root cause: the thermal compensation parameters of the robot controller had not been re-set for the 14 m track, and the joint between the second and third new-generation track sections was out of alignment by 0.08 mm. Corrective action: track joint re-shimmed and re-surveyed; compensation tables re-generated over the full 14 m; laser tracker volumetric check repeated. Preventive action: track survey added to the pre-FAT checklist (QRL-TP-117 rev 4).

7.2 Both NCRs were closed on re-test. No NCRs were raised on Cells 1 to 6.

8. Conclusion

8.1 All eight Cells have passed FAT in accordance with clause 8 and Schedule 1 Part C of the Agreement, and FAT Certificates FATC-01 to FATC-08 have been issued by the Buyer.

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8.2 Cells 1 to 6 passed at the first attempt. Cells 7 and 8 passed on re-test on 22/07/2024 and 29/07/2024 respectively.

9. Sign-off

Prepared by: Priya Sandhu, QA Engineer, Quellcrest Robotics Ltd - 02/08/2024

Reviewed by: Gary Whitlow, UK Project Director, Quellcrest Robotics Ltd - 02/08/2024

Approved by: Brenda Kowalczyk, VP Programs, Quellcrest Automation Inc. - 08/02/2024

Witnessed for the Buyer: Tomasz Wierzbicki, Quality Lead, Varnholt Aerostructures Ltd (all FATs)

Distribution: R. Thorne, T. Wierzbicki (Varnholt); G. Whitlow, K. Doyle, P. Sandhu (QRL); B. Kowalczyk (Quellcrest Automation Inc.)

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